

Multi-Domain Data Analysis Portfolio

5 Real-World Data Analysis Projects Across Distinct Domains

Author: Madhurja Bhattacharjee | **Program:** Developers Arena | Core Stack: Python 3.13, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Pytest

PROJECT 01

Supermarket Sales

Retail & Commerce

PROJECT 02

Student Performance

Academic & Education

PROJECT 03

Weather Analysis

Meteorology & Climate

PROJECT 04

COVID Trends

Healthcare & Epidemiology

PROJECT 05

Stock Market

Finance & Quantitative Risk

Project Overview & Analytics Methodology

Standardized 7-stage analytical pipeline applied consistently across 5 diverse domain datasets

STANDARDIZED WORKFLOW: 1. Ingestion -> 2. Cleaning -> 3. Validation & Testing -> 4. Statistical EDA -> 5. Visualization -> 6. Insights -> 7. Recommendations

Five Multi-Domain Workstreams

- Supermarket Sales: 2,000 retail transactions; branch performance, basket size, peak shopping hours, and ratings.
- Student Performance: 500 academic records; attendance %, study hours, subject scores, and performance tiers.
- Weather Analysis: 365 daily station records; annual rainfall concentration, temperature cycles, and extreme conditions.
- COVID-19 Healthcare: 1,825 regional tracking logs; case waves, vaccination phases, and regional mortality disparities.
- Stock Market Risk: 252 trading days; compounded return, price volatility, max drawdown, and Sharpe ratio.

Statistical Analysis & Metrics

- Central Tendency: Mean, median, and quartile analysis to detect skewness and avoid distortion from outliers.
- Dispersion & Risk: Standard deviation, IQR, daily and annualized volatility, and maximum drawdown.
- Correlation Analysis: Pearson correlation matrices with strict distinction between correlation and causation.
- Stratification & Tiers: Performance tier segmentation, customer classification, and vaccination phase stratification.
- Time-Series Profiling: 7-day moving averages, seasonal aggregation, and month-over-month compounding.

Tech Stack & Engineering Rigor

- Python Core Stack: Python 3.13, Pandas 2.2.0, NumPy 1.24.0, SciPy 1.12.0 for numerical computing.
- Visualization Engine: Matplotlib 3.8.0 and Seaborn 0.13.0 for accessible, labeled figures.
- Executive Reporting: FPDF2 2.7.8 for programmatic generation of formal PDF reports.
- Automated Testing: Pytest framework verifying shapes, nulls, duplicates, and formulas (18 tests passed).
- Integrity Principle: Zero artificial smoothing; raw source anomalies documented as limitations.

Supermarket Sales Analysis

2,000 retail transactions analyzed across 3 branches, 6 product categories, and 2023 seasonal patterns

Rs. 519,281 TOTAL REVENUE 2,000 Transactions	Rs. 259.64 AVG TRANSACTION Median: Rs. 258.04	6.96 / 10 AVG SATISFACTION Median: 6.95 / 10
---	--	---

Key Analytical Findings & Segment Profiles

- Branch Performance: Branch B leads at Rs. 180,821.79 (34.8%), Branch A closely tied at Rs. 179,047.93 (34.5%), while Branch C underperforms at Rs. 159,410.97 (30.7%, a 4.1% revenue gap).
- Afternoon Peak (12-5 PM): 42.2% of daily transactions (844 txns) occur in the afternoon rush, generating 9.7% more volume than morning (32.5%) and evening (25.4%).
- Product Line Balance: Health & Beauty leads at Rs. 96,970.32 (18.7%), followed by Sports & Travel (17.6%). The spread across all 6 lines is narrow (3.8%), indicating stable demand.
- Member Volume Dynamics: Members drive +Rs. 4,558 more total revenue via higher visit frequency (+36 txns), not basket size (Normal spends 1.86% more per txn: Rs. 262.08 vs Rs. 257.29).
- Gender Patterns: Female shoppers generated 3.68% higher total revenue (1,019 vs 981 txns) with higher satisfaction ratings (7.03 vs 6.88).

[DATA QUALITY LIMITATION: FINANCIAL CONSISTENCY]

The dataset contains a financial consistency discrepancy between the provided 'Total' column and 'Unit Price x Quantity + Tax' for most transactions. Original Total values were strictly retained without modification to preserve raw source data integrity.

Figure 1.1: Total Revenue and Transaction Volume by Branch

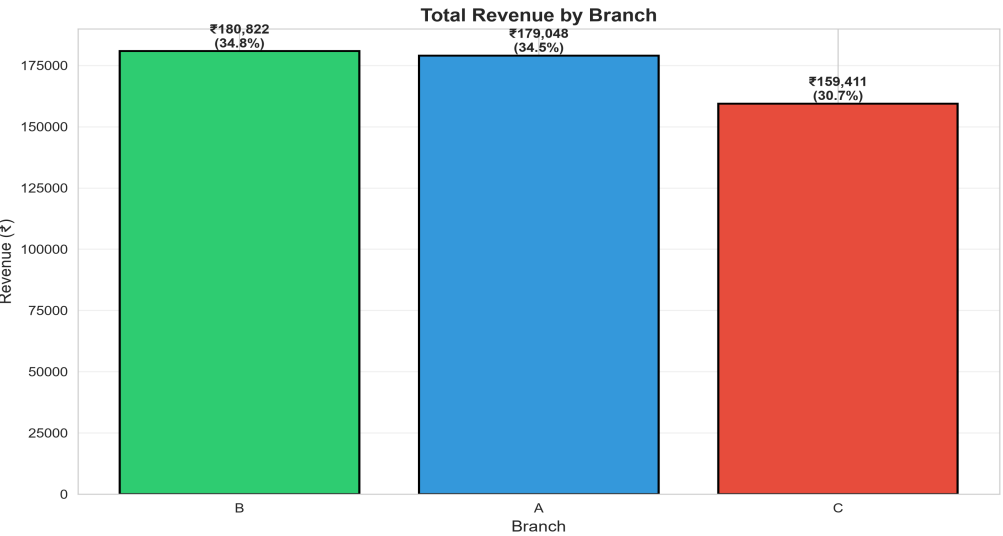
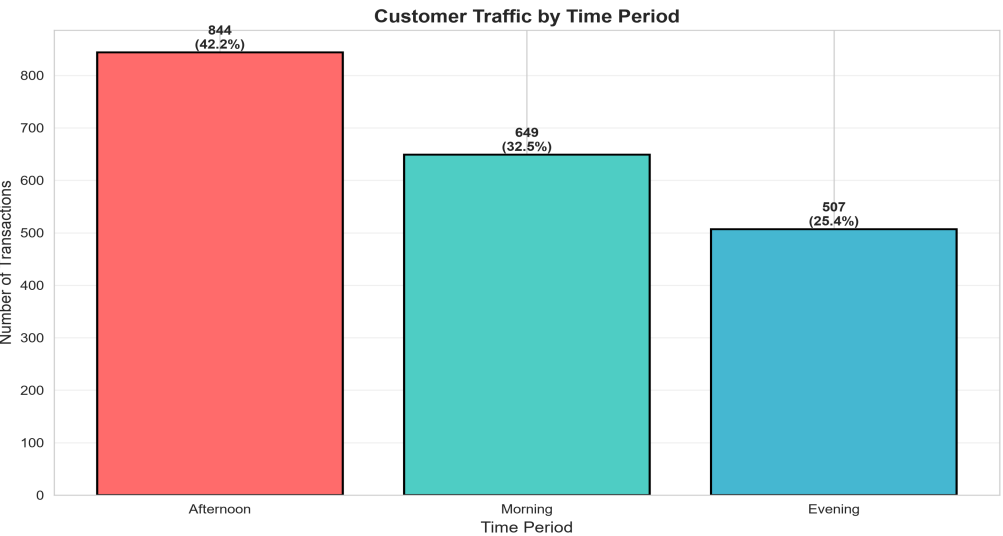


Figure 1.2: Hourly Shopping Distribution & Afternoon Peak Concentration



Student Performance Analysis

500 student records evaluated across 4 academic subjects, attendance rates, study hours, and performance tiers

74.38

AVERAGE SCORE

Median: 74.25 (100% Pass)

r = +0.077

ATTENDANCE CORR

Very Weak Linear Assoc

r = +0.096

STUDY HOURS CORR

Very Weak Linear Assoc

Key Analytical Findings & Segment Profiles

- Grade Distribution: Grade A: 27.4% (137 students), Grade B: 40.6% (203), Grade C: 25.8% (129), Grade D: 5.8% (29), Grade F: 0.4% (2). Overall pass rate was 100%.
- Subject Independence: Pairwise subject correlations are near zero ($r = -0.069$ to $+0.063$). Strong performance in Math does not linearly imply strong performance in Social Studies.
- Tier Disparities vs Effort: High Performers (27.4%) average 85.97, while Low Performers (6.2%) average 56.28. However, High vs Low attendance differs by only 2.97% (77.4% vs 74.4%) and study time by only 2.32 hrs/wk (24.2 vs 21.9 hrs).
- Subject-Wide Consistency: High performers score 84-88 across all four subjects, while low performers score 52-60 across all four, showing broad multi-subject separation.
- Actionable Takeaway: Mere attendance enforcement and counting study hours explain little outcome variation; interventions must focus on study quality and conceptual mastery.

[ANALYTICAL CAUTION: AVOID INNATE ABILITY / CAUSAL CLAIMS]
Weak correlations demonstrate that measured effort quantity explains little score variance. However, this observational dataset lacks variables on teaching quality, prior preparation, home environment, or study effectiveness. It does NOT prove innate ability or genetic causation.

Figure 2.1: Academic Grade Distribution Across 500 Students

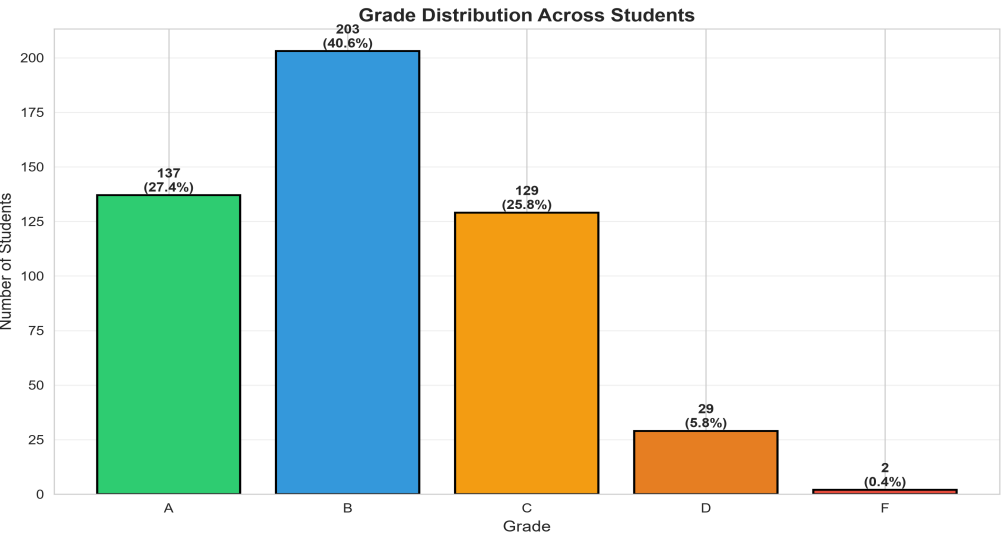
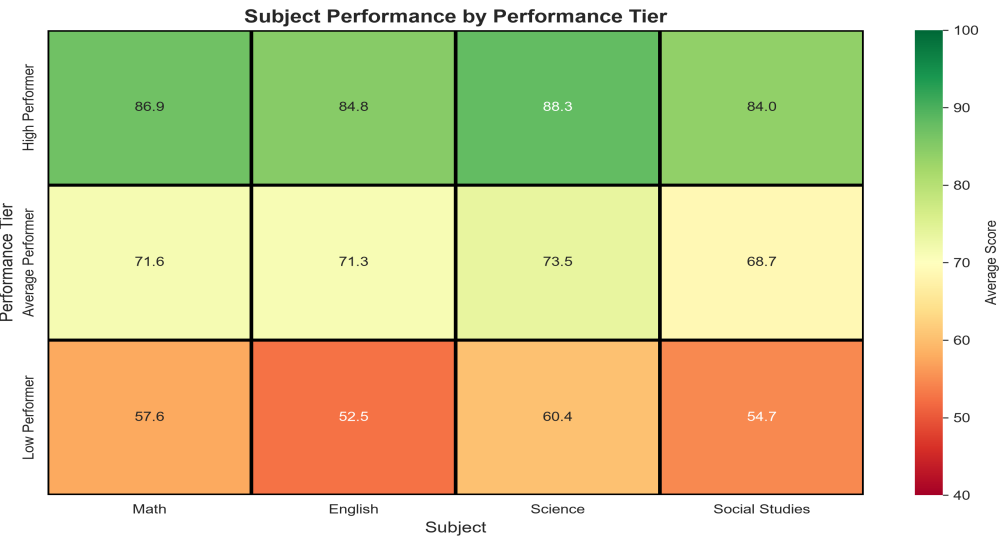


Figure 2.2: Subject Score Consistency Across Performance Tiers



Weather Data Analysis

365 daily observations (2023) from a single meteorological station analyzing temperature cycles and rainfall concentration

25.03 C

ANNUAL MEAN TEMP

Range: 0.6 C to 46.4 C

2,448 mm

ANNUAL RAINFALL

Daily Avg: 6.71 mm

64.7%

MONSOON SHARE

77 Heavy Days (>10mm)

Key Analytical Findings & Segment Profiles

- Seasonal Temperature Shifts: Spring is warmest (avg 38.07 C, max 46.4 C), Summer is moderate (avg 22.15 C), Winter is cool (avg 27.72 C), and Autumn is coldest (avg 12.10 C, min 0.6 C). Peak-to-trough spread is ~26 C.
- Monsoon Rainfall Concentration: 77 heavy-rain days (>10 mm) account for 1,583.00 mm (~65% of total annual rain), concentrated in June-September (June: 406.8mm, July: 328.5mm, Aug: 408.6mm, Sept: 408.9mm).
- Rainfall Frequency vs Intensity: Rain was recorded on 284 days (77.8% of year), but precipitation was heavily skewed: light rain days predominate, while acute monsoon bursts carry bulk volume.
- Condition Breakdown: Hot days: 119 (32.6%), Pleasant: 106 (29.0%), Rainy: 77 (21.1%), Cold: 63 (17.3%).
- Weak Cross-Variable Linear Correlation: Pairwise correlations between temperature, humidity, wind, and rainfall were weak ($|r| < 0.15$), reflecting non-linear atmospheric dynamics.

[SCIENTIFIC LIMITATION: SINGLE-STATION OBSERVATION]

This analysis strictly describes empirical observations from 365 days of 2023 data from a single station. A single year of data is insufficient to establish a formal long-term climate classification or multi-decadal trend under WMO standards.

Figure 3.1: Daily Temperature Trajectory and Extreme Bands Over 2023

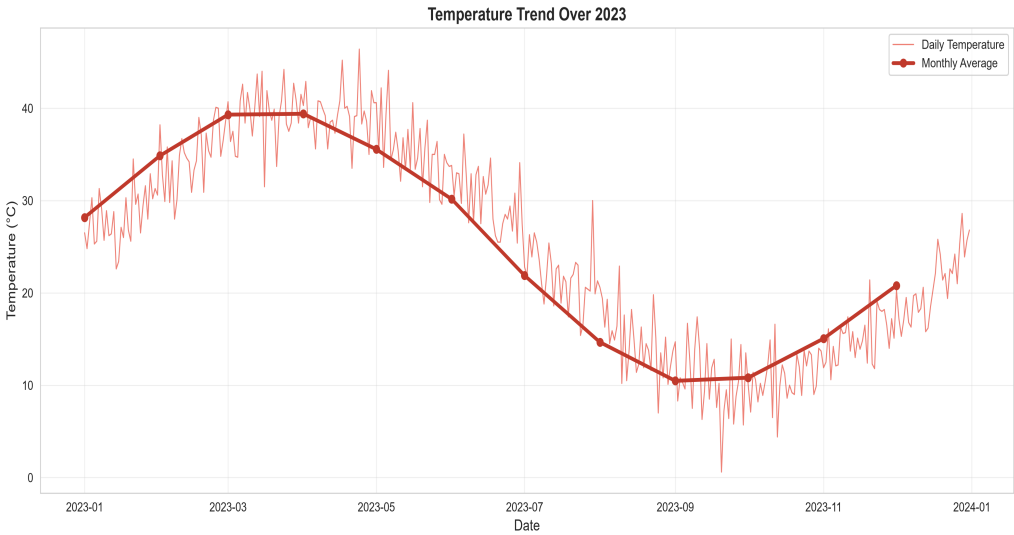
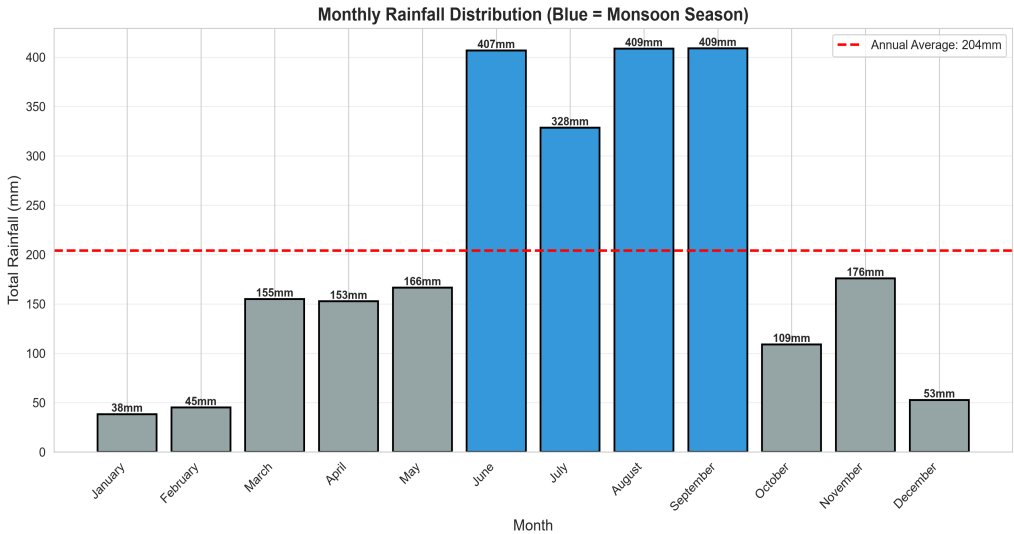


Figure 3.2: Monthly Precipitation Distribution Highlighting Monsoon Peak



COVID Trends Analysis

1,825 daily regional records across 5 geographic territories examining wave dynamics, mortality, and vaccination phases

391,062

TOTAL CASES

Deaths: 4,505 (CFR 1.15%)

86.23%

RECOVERY RATE

337,199 Recoveries

-56.7%

HIGH VAX PHASE

101 vs 234 Daily Cases

Key Analytical Findings & Segment Profiles

- Three Pandemic Wave Periods: Period 1 (Feb-Mar): Peak 57,108 cases in Feb. Period 2 (May-Jun): Annual peak 87,244 cases in May. Period 3 (Sept-Oct): Muted peak 35,377 cases in Oct. Dec low was 2,631.
- Vaccination Phase Association: High vaccination phases (60%+ coverage) averaged 101.28 daily cases vs 234.17 in low vaccination phases (0-30%), representing an observed 56.7% lower average daily case load.
- Regional Mortality Disparities: Central recorded 1,118 deaths (24.8% of total) vs East 787 deaths (17.5%). Central had 42.1% more deaths despite identical average vaccination rates (47.98%).
- Correlation Matrix Nuances: Vaccination rate vs Daily Cases showed moderate negative correlation ($r = -0.3574$); vaccination vs Daily Deaths was weak ($r = -0.1604$); Daily Cases vs Deaths was positive ($r = +0.3433$).
- Healthcare Capacity Implications: Mortality variation was substantially higher than vaccination differences, pointing to acute regional variations in ICU beds, hospital capacity, and age demographics.

[METHODOLOGICAL NOTE: DISTINGUISHING ASSOCIATION FROM CAUSATION]
The 56.7% case difference across vaccination phases is an observed statistical association. Observational aggregate data cannot establish singular clinical causation, as transmission was concurrently influenced by viral variants, public health mandates, and mobility.

Figure 4.1: COVID-19 7-Day Moving Average Trajectory Across 5 Regions

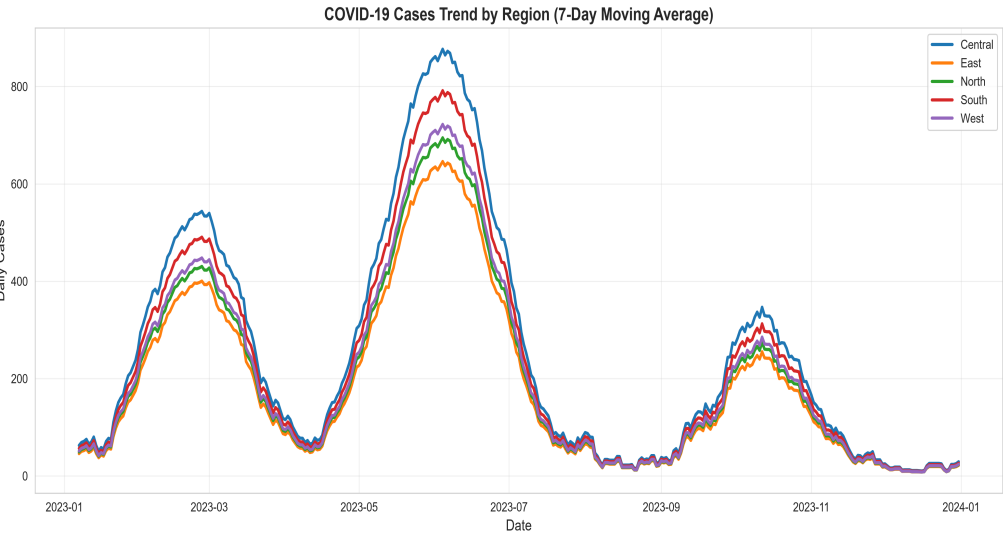
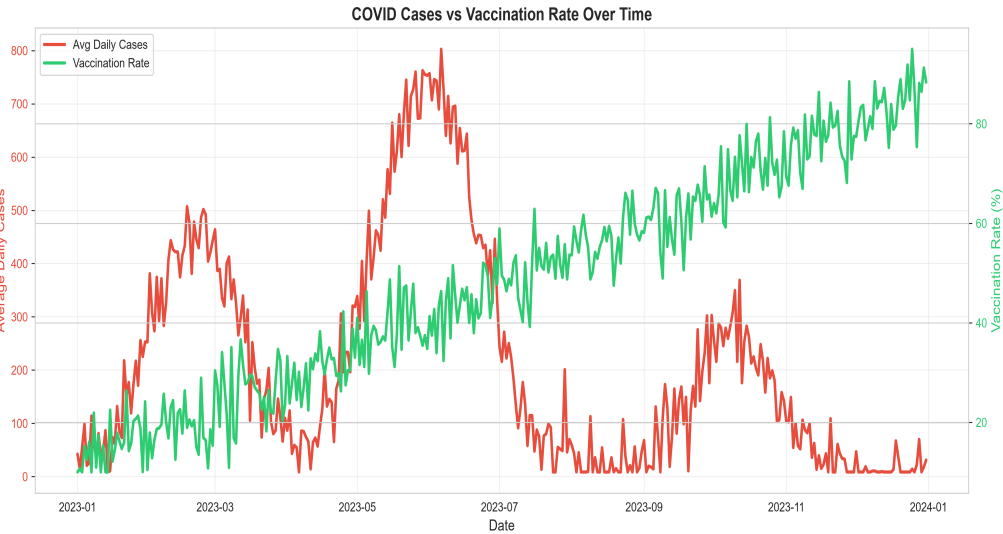


Figure 4.2: Vaccination Rollout Progress vs Daily Case Counts Over Time



Stock Market Analysis

252 trading days (Jan 2 to Dec 19, 2023) evaluating equity price trajectory, return volatility, and risk-adjusted metrics

+10.14% PERIOD RETURN Start \$101.05 -> End \$111.30	30.87% ANNUALIZED VOL Daily Volatility: 1.94%	-26.85% MAX DRAWDOWN Trough on May 26 (\$80.41)
---	---	---

Key Analytical Findings & Segment Profiles

- Price Range & Rebound: Lowest close reached \$80.41 on May 26; price subsequently rallied 46.4% to a peak close of \$117.74 on Nov 6, closing the period at \$111.30 (+10.14% return).
- Quarterly Trajectory: Severe early weakness in Q1 (-18.67% compounded, with Feb -9.79% worst month), followed by consistent rebound in Q2 (+8.89%), Q3 (+7.68%), and Q4 (+15.49%, led by Oct +17.91% best month).
- Daily Return Distribution: 130 positive days (51.6%) vs 121 negative days (48.0%). Daily returns ranged from -5.06% to +8.06%, with a median of +0.167% and mean of +0.057%.
- Sharpe Ratio (0% Rf Assumption): Annualized Sharpe ratio was 0.47, calculated using an explicit 0% risk-free rate assumption, reflecting modest risk-adjusted compensation for high 30.87% volatility.
- Volume-Return Disconnect: Correlation between trading volume and daily returns was $r = +0.017$, indicating volume spikes had negligible linear association with price direction.

[DATA QUALITY LIMITATION: OHLC INCONSISTENCIES]

The supplied dataset contains 78 rows with OHLC consistency issues: 40 rows where Open > High, and 38 rows where Open < Low. High-Low and Close relationships had 0 violations. Return formulas were validated (<1e-13% diff). Original data retained without alteration.

Figure 5.1: Daily Close Price Trajectory with 50-Day Moving Average

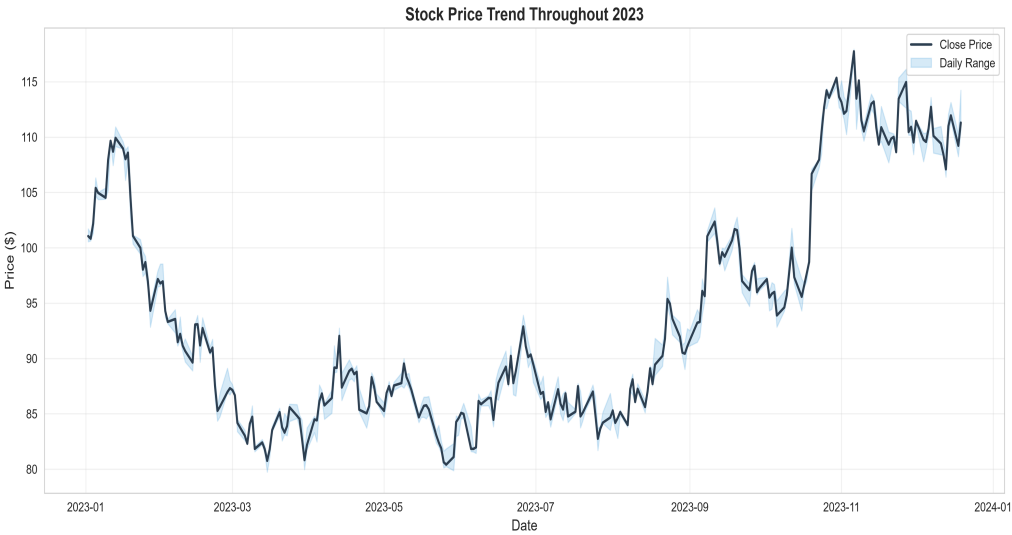
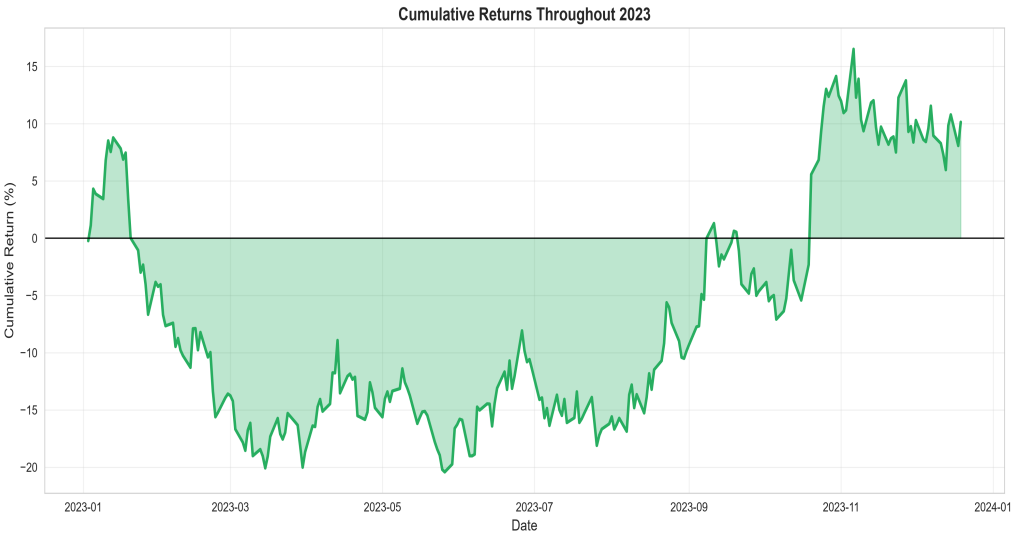


Figure 5.2: Cumulative Return Profile Showing May Drawdown and Recovery



Cross-Domain Insights & Analytical Comparison

Demonstrating methodological adaptability and statistical rigor across disparate business and scientific domains

Domain & Dataset	Core Analytical Question	Key Statistical Measures	Primary Visualizations	Decision Impact
01 Supermarket (2,000 txns, 14 cols)	What drives sales across branches, time periods, & customer segments?	Totals, branch shares, hourly volumes, mean/median ratings, spend parity	Grouped bars, distribution histograms, trend lines	Staffing shift to 12-5 PM peak; Branch C traffic investigation
02 Student Scores (500 students, 10 cols)	How do study time and attendance relate to grades and performance tiers?	Pearson r, tier averages, inter-subject correlation matrix, percentiles	Grade bar charts, subject heatmaps, effort grouped bars	Shift from counting study hours to assessing learning strategies
03 Weather Data (365 days, 12 cols)	What are the seasonal cycles, rainfall concentration, & extreme events?	Percentiles (10th/90th), annual precipitation, seasonal means & IQR	Time-series bands, monthly rainfall bars, seasonal boxplots	Rainwater storage for 65% monsoon volume; 86.4mm drainage design
04 COVID Healthcare (1,825 rows, 10 cols)	What are pandemic wave patterns, regional disparities, & vaccine phase trends?	7-day moving averages, CFR (1.15%), recovery rate (86.23%), phase means	Regional time series, dual-axis tracking, phase comparisons	Central region ICU capacity audit; targeted outreach for low-vax areas
05 Stock Market (252 trading days)	What are the asset return dynamics, price volatility, & risk-adjusted metrics?	Compounded return (10.14%), annualized vol (30.87%), drawdown, Sharpe	OHLC price trends, return histograms, drawdown profiles	Stress-test portfolios against -26.85% drawdown; reference \$80-\$85

Analytical Commonality Across Domains

- Structured Data Lifecycle: Each project advanced through uniform data ingestion, validation, and exploratory profiling.
- Integrity Preservation: Anomalies (OHLC inconsistencies, financial mismatches) were investigated and documented, not masked.

Turning Raw Data Into Decisions

- Avoiding Causal Traps: Maintained scientific discipline by treating weak correlations and associations without inventing causation.
- Testable Hypotheses: All operational proposals are framed as measurable, testable actions with trackable KPIs.

Data Quality, Validation & Automated Testing

Ensuring data integrity through automated unit testing, shape assertions, and anomaly detection

VERIFIED AUTOMATED TEST SUITE: 18 TESTS PASSED VIA PYTEST IN 1.36s

100% test pass rate across 6 test modules: test_supermarket.py (3), test_student.py (3), test_weather.py (3), test_covid.py (3), test_stock_market.py (3), test_data_validation.py (3).

Reusable Architecture (src/)

- `check_dataset_shape()`: Validates exact dimensional conformance against expected row and column counts.
- `check_missing_values()`: Audits global null counts across all dataset columns and data types.
- `check_duplicate_rows()`: Ensures zero duplicate record contamination across transactions and observations.
- **DRY Modular Design**: Standardized functions in `src/data_validation.py` imported directly into automated test scripts.
- **Defensive Engineering**: Enforces schema constraints before downstream statistical ingestion or modeling.

Dataset Dimensions & Completeness

- **Supermarket Sales**: 2,000 rows x 14 columns; 0 missing values; 0 duplicate rows.
- **Student Performance**: 500 rows x 10 columns; 0 missing values; 0 duplicate rows; verified score ranges (35-100).
- **Weather Observations**: 365 rows x 12 columns; 0 missing values; 0 duplicate rows; verified meteorological ranges.
- **COVID-19 Tracking**: 1,825 rows x 10 columns; 0 missing values; 0 duplicate rows; 5 regions x 365 days.
- **Stock Market Data**: 252 rows x 12 columns; 0 duplicate rows; 2 expected NaNs (Day 1 Daily & Cumulative Return).

Integrity Audits & Derived Metrics

- **Supermarket Financial Check**: Flagged calculation discrepancy between `Unit Price x Quantity + Tax` and `Total`. Original data retained.
- **Stock OHLC Validation**: Identified 78 rows with Open price anomalies (40 `Open > High`, 38 `Open < Low`). High `>=` Low had 0 violations.
- **Return Calculation Check**: Validated `Daily_Return` and `Cumulative_Return` against price formulas: `max error < 4.44e-14%` (floating point precision).
- **Preserving Raw Truth**: Adhered to industry compliance: anomalies documented as limitations rather than artificially altered.
- **Reproducible Pipeline**: Every check is version-controlled, automated, and re-executable in the CI/CD pipeline.

Cross-Domain Business Recommendations

Translating empirical data findings into testable operational adjustments, resource allocations, and monitored KPIs

01 SUPERMARKET SALES OPTIMIZATION

Target KPIs: Hourly queue times, Branch C revenue gap, low-rating %

Empirical Finding: 42.2% of transactions occur in the afternoon (12-5 PM); Branch C lags Branch B by 4.1%; 18.7% give low ratings.
Strategic Action: Reallocate staff & open additional checkout counters from 12-5 PM; audit Branch C footfall; address low-satisfaction issues.

02 STUDENT ACADEMIC SUPPORT

Target KPIs: Diagnostic assessment scores, percentage of students transitioning out of lower tiers

Empirical Finding: Study hours ($r=0.096$) and attendance ($r=0.077$) show very weak linear correlation with grades; effort between tiers differs by <3%.
Strategic Action: Move away from rigid study-hour mandates; implement diagnostic assessments early to identify specific concept gaps.

03 MUNICIPAL WATER & INFRASTRUCTURE PLANNING

Target KPIs: Rainwater capture volume, flood incident frequency during monsoon

Empirical Finding: 64.7% of annual rainfall (1,583 mm) falls in 77 days during June-September; peak daily rainfall reached 86.4 mm.
Strategic Action: Develop rainwater harvesting infrastructure before June; design storm drainage to 86.4 mm capacity; pause sensitive construction.

04 HEALTHCARE RESOURCE ALLOCATION

Target KPIs: Regional Case Fatality Rate (CFR), ICU bed occupancy rate, localized vaccine coverage

Empirical Finding: Central region recorded 42.1% excess mortality over East despite identical average vaccination rates (47.98%).
Strategic Action: Prioritize clinical ICU and ventilator resources to Central region; deploy targeted mobile vaccine clinics to low-coverage pockets.

05 QUANTITATIVE RISK MANAGEMENT

Target KPIs: Trailing volatility, maximum peak-to-trough drawdown, benchmark-adjusted Sharpe

Empirical Finding: Stock delivered +10.14% return but suffered -26.85% maximum drawdown and 30.87% annualized volatility (Sharpe 0.47).
Strategic Action: Treat \$80-\$85 as historical reference rather than guaranteed support; implement risk management rules to limit drawdowns.

Technical Implementation & Repository Structure

Modular, reproducible data science architecture adhering to production software engineering best practices

Production Data Science Stack

- Python 3.13: Modern runtime executing data manipulation and analytics packages.
- Pandas & NumPy: Fast, vectorized dataframe manipulation, aggregation, and mathematical computations.
- SciPy: Scientific statistical distributions, descriptive calculations, and correlation metrics.
- Matplotlib & Seaborn: Production visualization engine generating 22 high-resolution, labeled figures.
- FPDF2: Programmatic generation of 5 formal executive analytical PDF reports.
- Pytest: Automated unit test suite verifying schema shapes, missing data, and financial formulas.

Engineering & Reproducibility Principles

- DRY Architecture: Reusable validation routines centralized in src/data_validation.py across all five domain projects.
- Defensive Quality Checks: Automated pre-flight assertions for schema dimensions, null values, and duplicate detection.
- Reproducible Artifacts: Automated scripts compile raw data into notebooks, formal PDF reports, and decks.
- Strict Data Provenance: Raw data is never altered in-place; source inconsistencies are documented transparently.

ORGANIZED REPOSITORY STRUCTURE

```
Multi-Domain-Data-Analysis/  
|-- data/raw/                # 5 raw CSV datasets  
|-- notebooks/               # 5 production Jupyter notebooks  
|   |-- 01_supermarket_sales_analysis.ipynb  
|   |-- 02_student_performance_analysis.ipynb  
|   |-- 03_weather_data_analysis.ipynb  
|   |-- 04_covid_healthcare_analysis.ipynb  
|   \-- 05_stock_market_analysis.ipynb  
|-- src/                     # Reusable data validation module  
|   \-- data_validation.py    # Shape, null, and duplicate check functions  
|-- tests/                   # Automated pytest test suite (18 passed)  
|   |-- test_data_validation.py  
|   |-- test_supermarket.py / test_student.py  
|   \-- test_weather.py / test_covid.py / test_stock_market.py  
|-- visualizations/          # 22 high-resolution saved charts  
|-- reports/                 # 5 formal executive analytical PDF reports  
|-- presentation/            # PowerPoint (.pptx) & interactive HTML  
|-- docs/                    # Documentation & verification logs  
\-- requirements.txt          # Pinned dependency manifest
```

Conclusion: Turning Data into Decisions

Demonstrating end-to-end data analytics capability, technical rigor, and objective business insight generation

1. Data Ingestion & Cleaning

Standardized handling of diverse data formats and dates, feature engineering (time-periods, moving averages), and structured data hygiene across all datasets.

2. Validation & Auditing

Pre-flight shape assertions, missing value verification, duplicate detection, and explicit documentation of raw data anomalies without artificial distortion.

3. Statistical EDA

Rigorous application of central tendency, dispersion metrics, Pearson correlation matrices, and distribution profiling tailored to each domain.

4. Publication Visualizations

22 production-grade Matplotlib/Seaborn figures with consistent branding, readable labels, units, and clear analytical value.

5. Business Strategy

Translating empirical patterns into testable operational adjustments while strictly avoiding causal claims from observational associations.

6. Testing & Engineering

Automated pytest test suite with 18 passed tests, modular src/ architecture, pinned dependencies, and reproducible PDF reporting.

"TURNING DATA INTO DECISIONS"